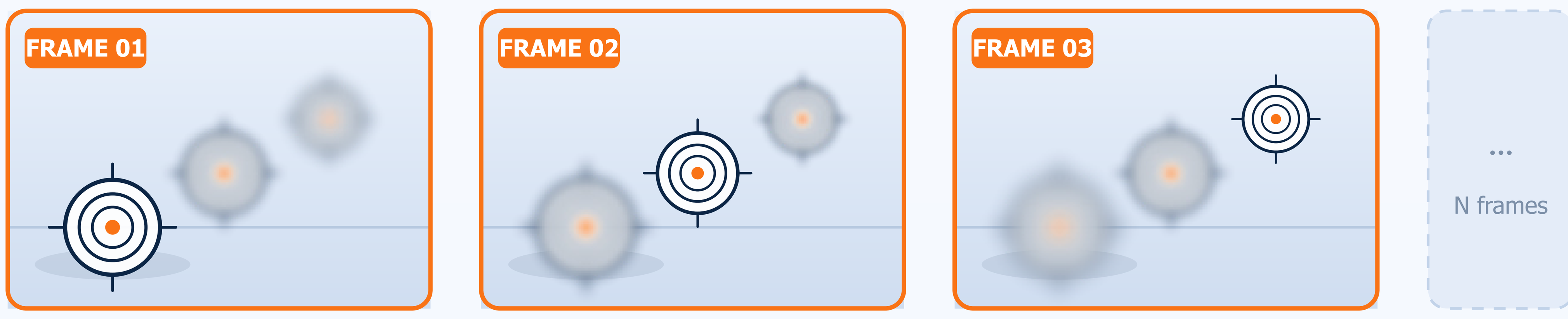
DIGITAL PHOTOGRAPHY · MULTI-FOCUS FUSION

# Focus Stacking with the Laplacian Pyramid

Burt & Adelson (1983) decomposition · energy-weighted band selection · one all-in-focus render

## 1 Capture the stack

Each frame is taken at a different focus distance. At macro magnification only a thin slab of the subject resolves.




focus @ near plane · focus @ mid plane · focus @ far plane

**Why one exposure is not enough**

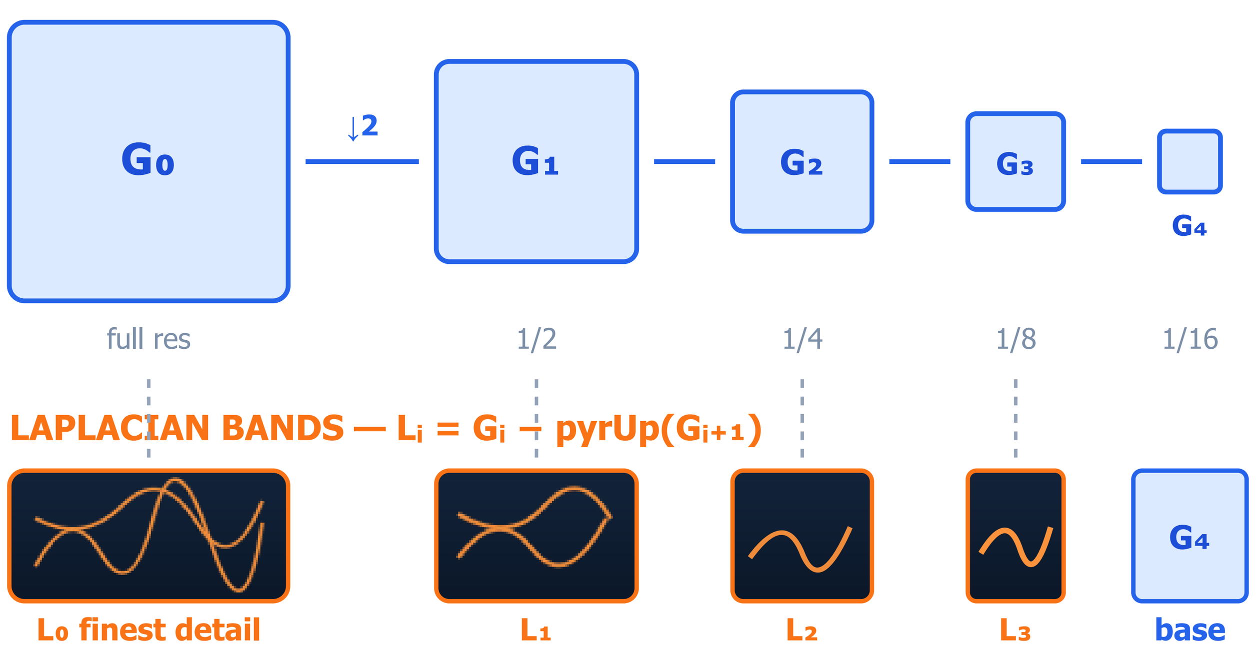
Depth of field shrinks with magnification. At 1:1 and f/8 it can fall below a millimetre, so no single aperture renders the whole subject — stopping down further trades focus for diffraction blur.

**Solution: fuse many frames, not one.**

## 2 Decompose — build the Laplacian pyramid

Every frame is split into scales: repeated 2×-downsampling gives a Gaussian pyramid; each band is what one level loses to the next.

**GAUSSIAN PYRAMID — pyrDown, blur + decimate**




**The pyramid is lossless — and that is the point**

The bands add back exactly to the original image, so anything decided band by band can be reassembled without inventing pixels.

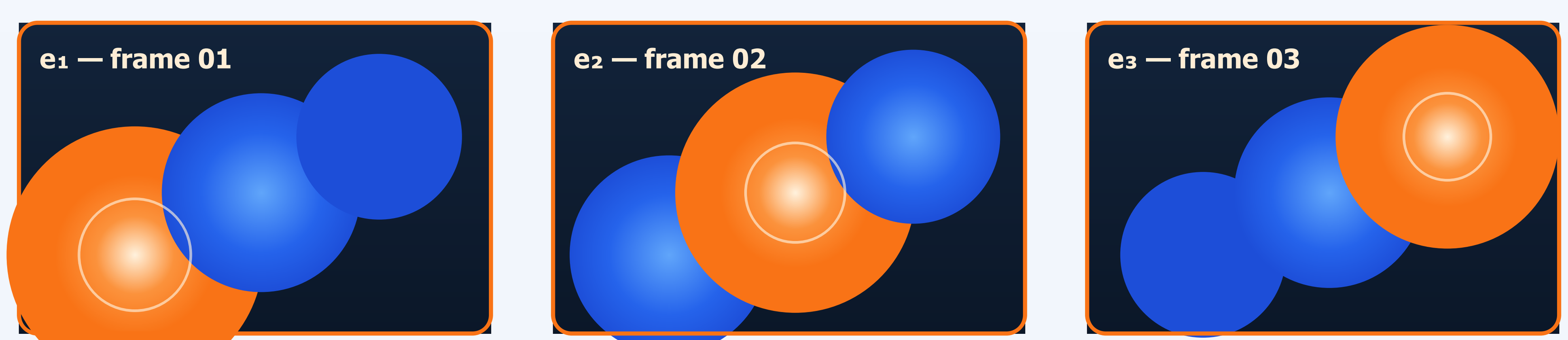
**Each band answers a different question:**

- $L_0$–$L_1$ — pore-level texture, grain, the finest hairs
- $L_2$–$L_3$ — contours, structural edges, coarse relief
- $G_4$ base — overall tone and colour, no detail at all

Working across scales is what stops a single sharp speck from dragging a whole region with it.

## 3 Measure focus — local energy per band

A single |Laplacian| pixel is noise. Pooling its square over a 5×5 window turns it into a region measure: a focus heatmap per frame, per band.




near plane wins here · mid plane wins here · far plane wins here

no detail resolved — defocus, flat tone, grain only · high band energy — genuinely in focus

**Focus energy, normalised**

$$e_i = \mathrm{box}_{5\times5}(\, L_i^2 \,) \,/\, n_i$$

Dividing by $n_i$, the frame's own noise level (10th percentile of its live band energies), stops a bright defocused veil from outscoring a dark sharp frame on grain alone. Every frame then reads about 1.0 where it resolves nothing — so those regions tie.

## 4 Select & merge — band by band, pixel by pixel

The classic rule copies the strongest coefficient. A soft exponent keeps that decision where it is real and averages where nothing is in focus.

**Weighting rule**

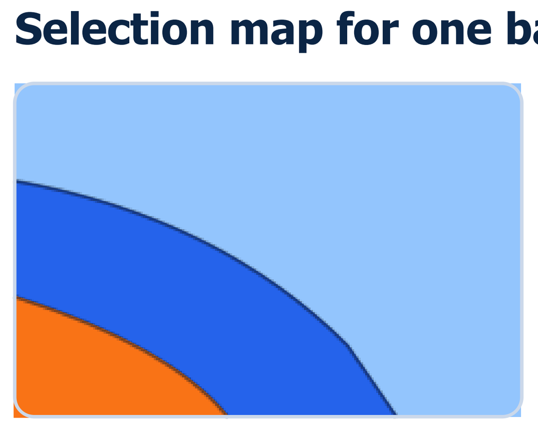
$$w_i = (\, e_i \,/\, e_{max} \,)^{\beta} \text{ with } \beta = 8$$

CONTRIBUTION TO THE FUSED BAND

| | |
|---|---|
| frame 02 | 100 % |
| frame 01 | 4 % |
| frame 03 | 0.4 % |

A frame 2× behind the winner contributes $2^{-8} \approx 0.4$ % — effectively a decision, without the speckle.

**Selection map for one band**



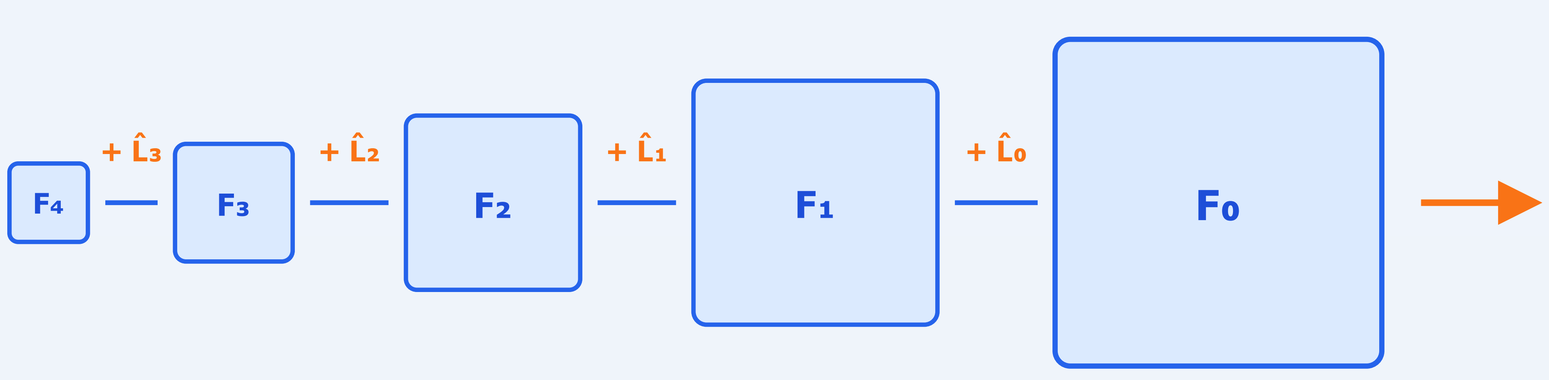
- frame 01 carries this region
- frame 02 carries this region
- frame 03 carries this region

**Boundaries follow depth, not a grid.**

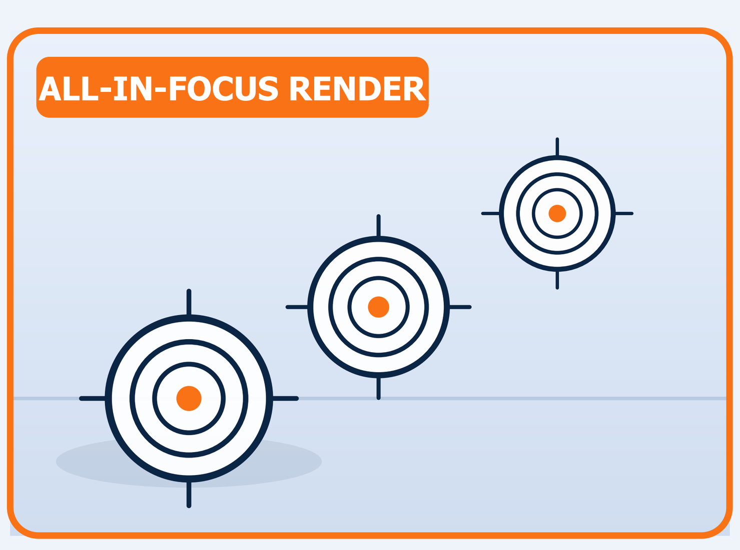
Where frames tie, the weights average them — the correct answer for “nothing resolves here”.

## 5 Collapse the pyramid — one fully focused image

From the coarsest level down: upsample, add the fused band, repeat. The stack is gone; the detail of every frame is not.




$F_i = \hat{L}_i + \mathrm{pyrUp}(F_{i+1})$ — coarsest level first, fused band added at every scale




**Front to back sharp — from a stack no single exposure could capture.**

**PIPELINE AT A GLANCE**

align → decompose → score → weight → merge → collapse

| | |
|---|---|
| levels | 5 |
| pooling window | 5 × 5 |
| selectivity β | 8.0 |


Burt P. J., Adelson E. H. — “The Laplacian pyramid as a compact image code”, IEEE Trans. Communications 31(4), 1983.